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| FLASH GSM                           |            | WORK INSTRUCTIONS |              |
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# Nokia 3310 manual

## Introduction

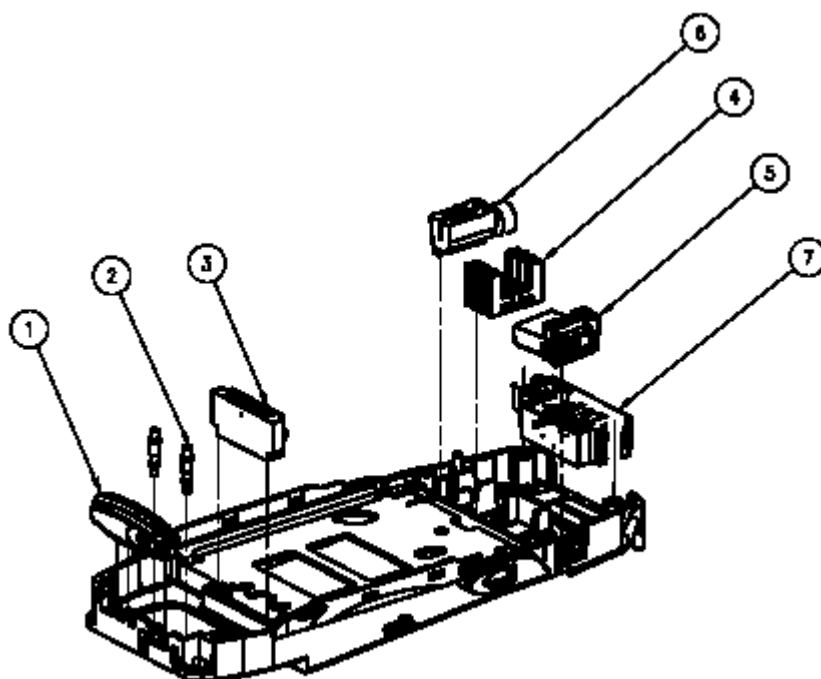
This manual is intended to help you carry out repairs on level 2 and 3 meaning limited component repairs and full lab repairs. It must be noted that all repairs have to be carried out in an ESD environment (electrostatic sensitive devices).

As this is a manual based on knowledge and experience any update is welcomed. For update & improvements please contact Seni Dragos at Repair department.

## Level 2

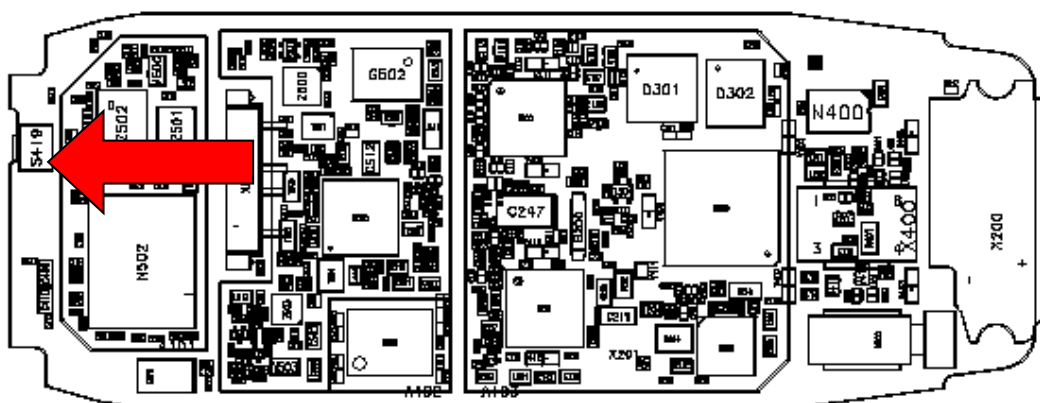
- 1)Cover power knob+power button
- 2)Antenna connectors+antenna
- 3)Battery connector
- 4)Sim connector+SIM power supply
- 5)Buzzer+buzzer driver
- 6)Vibrator+vibrator driver
- 7)Charging & microphone connector
- 8)LED-s and keyboard

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## Chapter 1-Cover power knob+power button

- 1)Faults-The phone don't power up
- 2)Location:Nr 1 in the schematics and the arrow under



3)Power knob part number.....

Power button part number.....

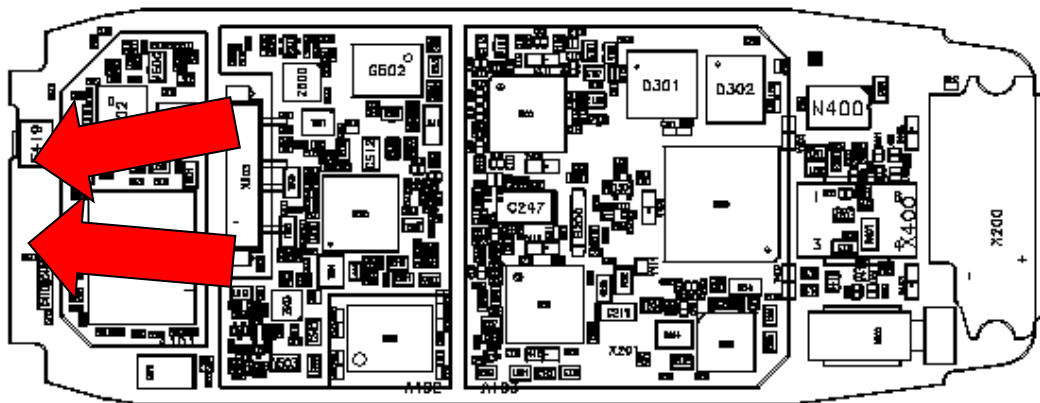
4)Diagnosis-Put the multimetre on diode check and when you press the button the extremities will have continuity.If not is faulty will not be continuity.The power button cover could be broken.

5)Repair by component change-Use the hot air to remove the component.Use around movements to not melt the plastic from the button.

## Chapter 2-Antenna connectors+antenna

1)Fault symptoms-Phone have no network or have problems entering network

2)Location on board-nr.2 and the arrows below



3)Antenna connectors part number.....

Antenna part number.....

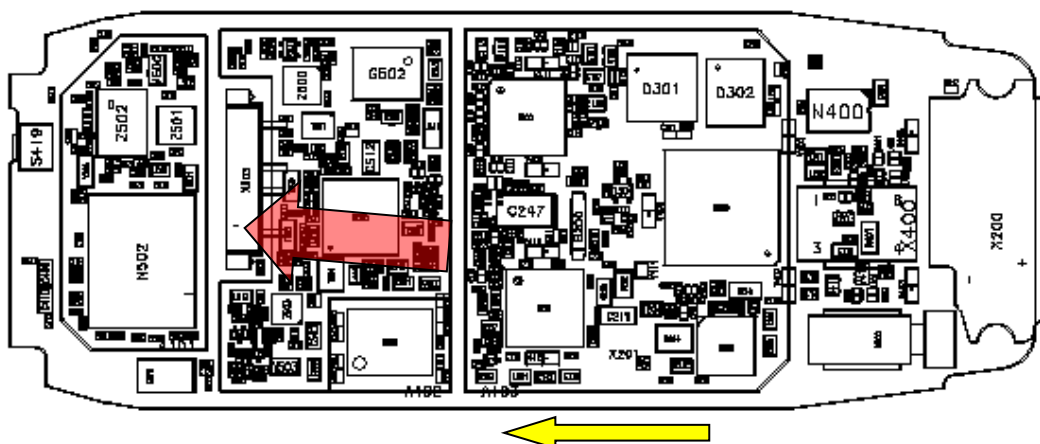
4)Diagnosis-Check the antenna and connectors for oxide and for bad contacts.Check the path between antenna connector and RXTX switch (Z502).

5)Repair by component change-Use a standard screwdriver to remove the antenna and the antenna connectors.

### Chapter 3-battery connector

1)Fault symptoms-Phone doesn't power up or the phone display"not charging" when a charger is connected

2)Location –nr 3 in the schematics and the arrow below



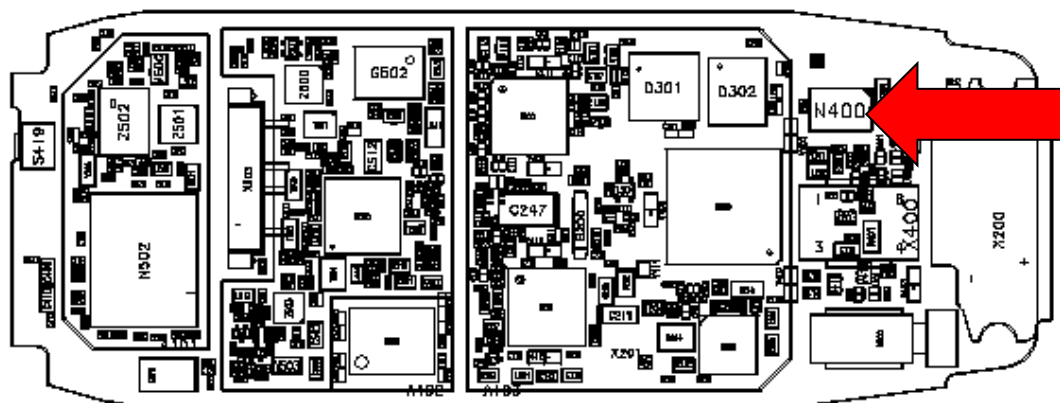
3)Battery connector part number.....



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## Chapter 5-Vibrator+vibra driver

- 1)Fault symptom-No vibration
- 2)Location-fig.5+Arrow below



3)Vibrator part number.....

Vibra driver part number.....

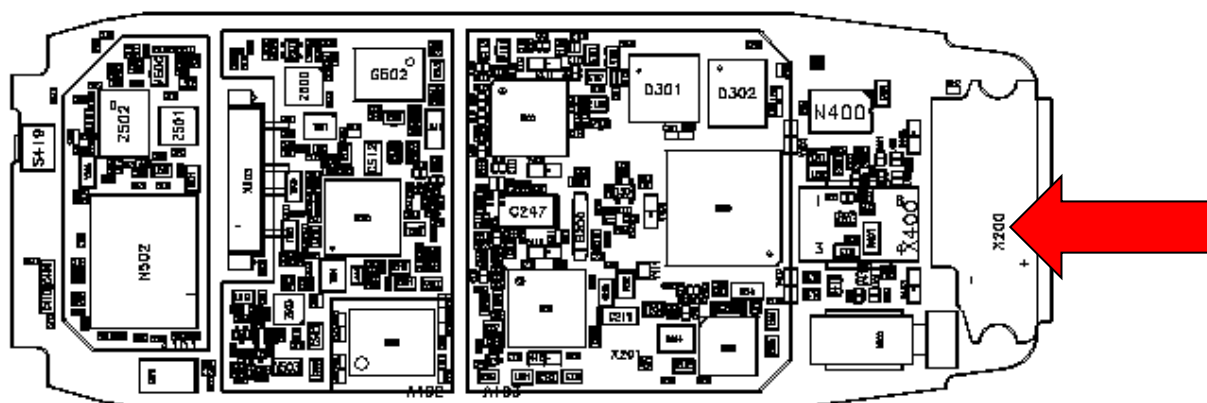
4)Diagnosis-Take the vibrator out of the case and put its terminals at 2v.If the vibra-motor rotates means the vibrator motor works. Check the paths between Vibrator pads and Vibra-driver N400.

5)Repair by component change-Use the hot air blower to remove and change the defective N400.

## Chapter 6-Buzzer+buzzer driver

- 1)Fault symptoms-No ringing
- 2)Location-nr5 +the arrow below(driver)





3)Part number bottom connector.....

part number microphone.....

4)Diagnosis-Take out of the case the connector and make am carefull oxide verification.Connect a charger at the connector and verify the voltage at the other end (3.7v).Verify the contacts on the microphone.Use for cleaning Kontakt 60 and alcohol.If oxide is present use only Kontakt 60.

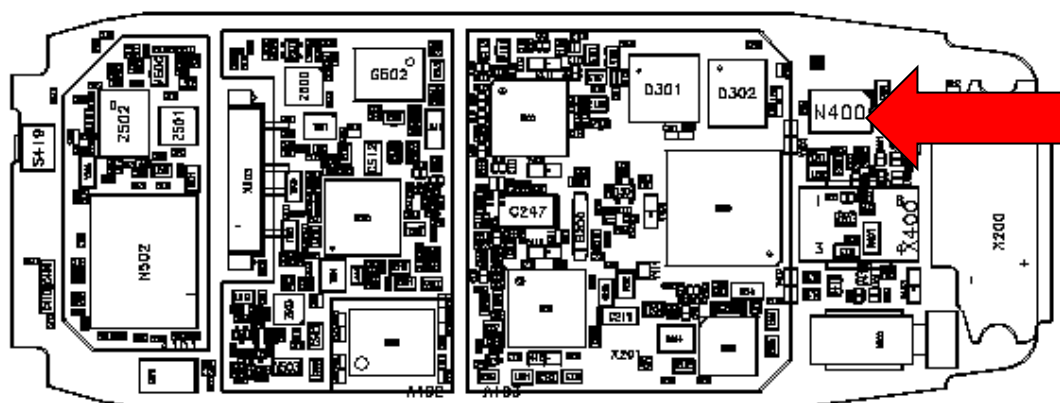
5)Repair by component change-Change manually the microphone or the connector.

## Chapter 8-LED-s and keyboard

1)Fault symptoms-No light totally or partially.The keys don't work all or culomns,lines ,single.

2)Location-On the front of the PCB.

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- 3)Parts number LED-s.....
- 4)Diagnosis-80% of the faults are because of interruptions in the paths. Keypads are organized in a matrix in columns and lines. Verify with a multimeter the paths. The LED-s are alimented from N400.If none of them works probably the IC is faulty. If only some of them works probably some paths are interrupted.
- 5)Repair by component change-Use the hot air blower to change the IC-s. Use the hot air blower carefully in round movements in order not to melt the LED-s. **Watch the polarity!!!.**

## Level 3

Level 3 in repairings means full laboratory repairs.In order to perform theese repairings you need to have experienced technical skills.You need to practice repairings assisted by an experienced technician at least 6 month.Also you need to have advanced knowledge in electronics.This manual was made in order to give you information that normally is achived in months.This is not a Service factory manual

because it will not tell you detailed description of the circuit diagram.For that you can consult service factory manuals and please feel free to do so.But if you want to know the service problems that our



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Diplex filter parts number.....

4)Diagnosis-When a phone has network problems the most suspected component is the P.A.

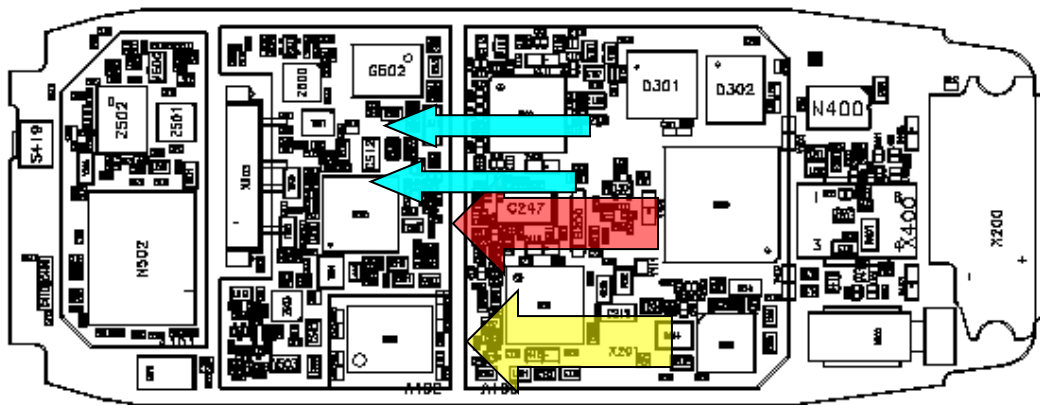
because it is a component that in use has power full currents and often in conversation it is heating. It is mostly responsible for not being able to making a call. The diplex filter is very sensible on shocks and oxide and is responsible of the phone not entering the network.

5)Repair by component change-Use the hot air blower in circular movements and take the PA and the diplex filter from the ceramic base not from the metallic shield. Use sharp enough tweezers.

## Chapter B-Rx,Tx coupler,VCO,Haggar

1.Fault symptoms-No network(Rx coupler,VCO,Haggar),Tel can't make the call(Tx coupler ,Haggar)

2.Location on board-Blue arrows Rx,Tx coupler,Red arrow Haggar(RF I.C.),Yellow arrow VCO



3.Rx,Tx coupler part number.....

VCO part number.....

Haggar part number.....

4.Diagnosis-First put the tel. on power supply alimentation .If the phone consumes more then 120mA then then one component in the

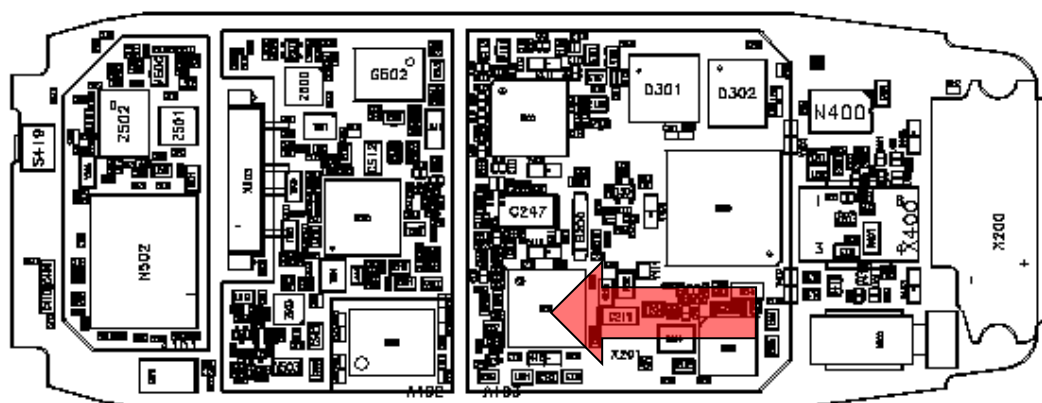
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phone is leaky and in the network section couldn't be other but the P.A. On Nokia networks problems are always very difficult to repair because in the network detection and emission process are involved 75% of the components .Imagine that one pin at the GSM processor is dry soldered and the phone doesn't "see" the network .That phone could't be repaired. In order to perform measures with the oscilloscope or other equipment to see where the signal pass is interrupted you need to have the special TEST console from NOKIA. Even though to perform these measures is not an economical repair but only for series of phones that have the same fault(very rare situation at Nokia).So you have to remember that if you can't repair the phone in 10 minutes you have to put it at BER phones. So this is an example: You have to repair a phone that drops network

5)Repair by component change-First you have to heat the board carefully with the hot air blower to be sure that the components have no dry soldering joints .After that change the P.A and if the fail persist change in order: Diplex filter,Cobba,26 Mhz clock,Real time clock,Rx Tx coupler,Haggar,VCO and if the fail persists could be other components related with the Network components or interrupted main board.

## Chapter C-CCONT

- 1.Fault symptoms-The phone doesn't power up ,Not charging and even network problems
- 2.Location on board



3.Part number.....

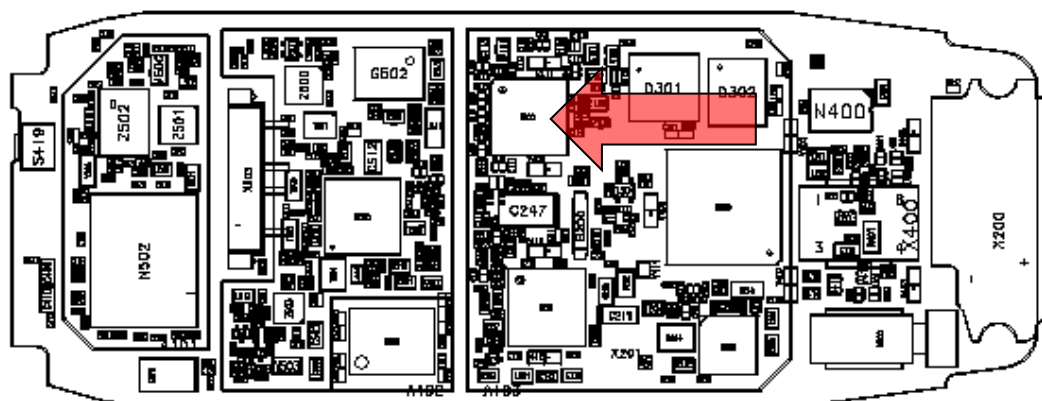
4.Diagnosis-If the phone doesn't power up after you check the battery connector,the software of the phone,the path between battery connector and the CCONT you have to change the CCONT.

For a safe work with the BGA component use a kit for mounting BGA and a sharp tweezers.

## Chapter D-COBBA

1.Fault symptom-Network problems, the microphone and the speaker doesn't work in conversation, "contact service"

2.Location on board



3.Part number.....

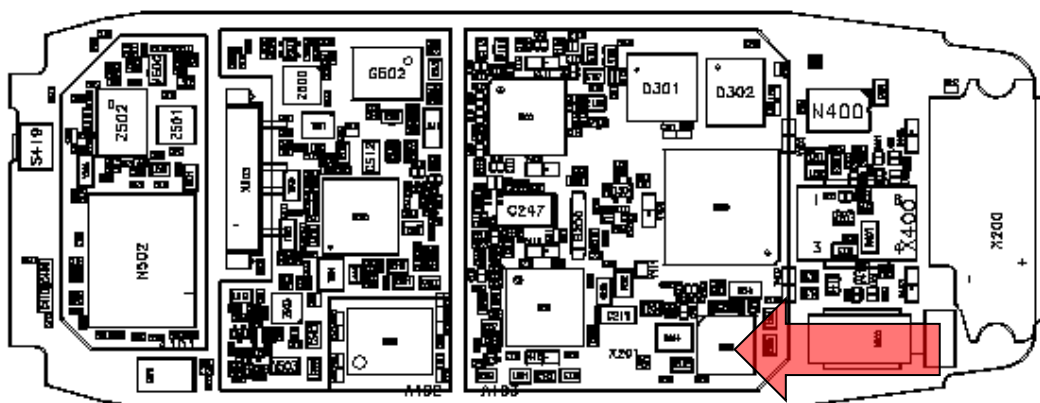
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4.Diagnosis-This is the second component to check after P.A. and Diplex filter if the phone have network problems.If the phone have microphone problems (ex the phone doesn't records or the keypads sound "tone"doesn't work it means that the CCOBA is not faulty but the microphone or the speaker itself. If the microphone or the speaker doesn't work in conversation it is a strong possibility that the COBBA is faulty. First you heat it with the hot air blower to be sure that there are no dry soldering joints and after that you change it. Use a kit for mounting BGA components if you use components recovered from BER phones.

## Chapter E-Charging I.C.

1.Fault symptom-When charger is inserted phone announce "NOT CHARGING"

2.Location on board-



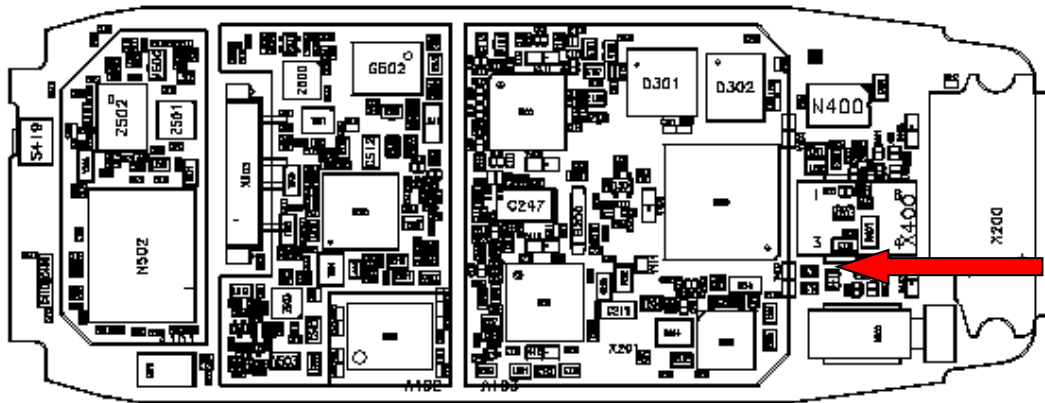
3.Part number.....

4.Diagnosis-Before trying to change or to solder the I.C. verify the battery connector if the middle connectors are not interrupted .Pay attention because this I.C. is near the GSM processor and the heat could damage it because it has a special raisin below it.

## Chapter F-Special reports

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1. The first special report is about a fault that is very usual to 4.24 version of 3310. You power up the phone with the card inside. After an interval between 1 min and 5 min the phone will block. The keypad doesn't work so in order to power off the phone you have to remove the battery. You have to change a resistor near the charging I.C.. The component is located in the picture below



This resistor is R306 CLK TEST and you have to change it with a resistor around 50 Ohm that you can find in Nokia 5110

16.04.2003

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